

CropSage Crop Catalog Schema

Human-readable companion to catalog.schema.json. The JSON Schema remains the machine validation contract.

Required-field rule: every crop record contains all 29 finalized fields. Version 1.1.0 adds effective_root_zone_depth_cm.

Schema Overview

Property	Value	Meaning
Schema draft	https://json-schema.org/draft/2020-12/schema	JSON Schema dialect used by the machine contract.
Schema ID	https://cropsage.local/schemas/crop-catalog-1.1.0.json	Stable identifier for this schema version.
Catalog version	1.1.0	The schema accepts catalog version 1.1.0 only.
Crop count	exactly 22	The release contains the selected Texas crop profiles.
Required crop fields	29	Additional crop properties are rejected.

Finalized Crop Fields

Group	Field	Type	Stored requirement	Location comparison / runtime use
Identity	crop_id	string	Stable unique snake_case identifier.	Catalog lookup and audit key.
Identity	common_name	string	Farmer-facing crop profile name.	Displayed in rankings and explanations.
Identity	scientific_name	string	Botanical taxon for the profile.	Prevents common-name ambiguity.
Identity	production_use	string	Harvest endpoint such as grain, silage, hay, lint, or fresh market.	Separates profiles such as grain corn and silage corn.
Identity	variety_scope	string	Included cultivar or market classes and explicit exclusions.	Flags assumptions before scoring.
Region and season	supported_texas_regions	region_id[]	Texas regions where the crop can enter ranking.	Determines whether the profile is supported at the location.
Region and season	regional_suitability	object[5]	Rating and basis for every catalog region.	Provides regional prior and explanation.
Region and season	planting_windows_by_region	object[]	Month-level windows, evidence status, and sources.	Checks whether the date is inside a supported window.
Region and season	days_to_maturity	object	Minimum and maximum calendar days with basis and evidence.	Checks whether the crop can mature in the available season.
Temperature	optimal_temperature_range	object	Minimum and maximum degrees C plus measurement basis and sources.	Measures temperature fit.
Temperature	heat_stress_threshold	object	Optional warning threshold, operator, severity, and scoring permission.	Creates sourced heat warnings or soft penalties.
Temperature	heat_exposure_duration	object	Optional hours or days; null when not established.	Controls whether a heat event may create a penalty.
Temperature	heat_sensitive_stages	string[]	Growth stages most vulnerable to heat.	Applies heat evidence at the relevant stage.
Temperature	frost_sensitivity	object	Frost class, explanation, and sources.	Flags establishment or maturity frost risk.
Temperature	temperature_unit	constant	Always degC.	Prevents conversion ambiguity.
Temperature	temperature_measurement_basis	string	Air, soil, canopy, daily maximum, or other stated basis.	Prevents comparison of unlike measurements.
Soil	preferred_soil_textures	string[]	Preferred agronomic texture classes.	Produces a soft texture-fit result.

Group	Field	Type	Stored requirement	Location comparison / runtime use
Soil	ph_tolerable_range	object	Minimum and maximum tolerated pH with evidence and sources.	Checks whether mapped or measured pH is within range.
Soil	effective_root_zone_depth_cm	object	Sourced mature-crop min-max effective depth; nullable reference value.	Caps horizon water integration and produces an accessible-storage range.
Soil	drainage_requirement	object	Drainage class, explanation, and sources.	Flags likely drainage mismatch.
Water	water_demand	object	Demand class and optional seasonal millimetre range.	Compares demand with water evidence without treating storage as supply.
Water	drought_tolerance	enum	Very low through very high tolerance class.	Adds crop resilience context.
Water	irrigation_requirement	object	Usually required, recommended, conditional, or often rainfed.	Filters or warns when required infrastructure is unavailable.
Evidence	source_ids	string[]	References supporting the complete record.	Allows evidence tracing and validation.
Evidence	confidence	enum	High, medium, or low record-level confidence.	Contributes to recommendation confidence, not suitability.
Evidence	last_reviewed	date	Most recent catalog review date.	Flags stale records.
Evidence	assumptions	string[]	Explicit scope, transfer, and variety assumptions.	Exposes limitations and possible conflicts.
Evidence	catalog_version	constant	Version 1.1.0 for this release.	Makes recommendations reproducible.
Evidence	record_status	enum	Provisional, approved, or retired.	Prevents retired or unreviewed records from silent production use.

Root-zone Object Contract

Child field	Contract	Use
min	Integer centimetres, 1-500	Lower published effective-depth bound.
max	Integer centimetres, 1-500; must be at least min	Upper published effective-depth bound.
reference	Integer in range or null	Only populated when a source supports one representative value.
basis	Non-empty explanation	Defines mature-crop and unrestricted-soil semantics plus transfer assumptions.
evidence_status	Catalog evidence enum	Controls whether the range can support soft scoring.
source_ids	At least one catalog reference	Links the numeric range to FAO-56 and regional context where available.